

Business Analyst Case Study

Context

We run a B2B platform that helps small merchants ship products. Marketing brings in inbound leads, our Filtration team calls each lead to qualify them, and qualified leads get handed off to a Sales team through a fixed buddy pairing. A conversion is when a merchant books their first shipment with us. Everything below sits on top of that flow.

The data

You'll get one Excel file (**BA_Assignment_Data.xlsx**) with four tabs.

Tab 1 — Team Legend

A map of every agent code that appears in the other tabs, showing which team they belong to and (for Filtration agents) the Sales agent they hand qualified leads to.

Tab 2 — Lead Data

One row per lead created in the last 30 days. This is the current state of each lead.

Lead ID	Unique identifier (Lead_001 to Lead_100)
Signup Date	When the lead was created in the system
Owner	Agent code currently owning the lead
Contact Stage	Current stage in the pipeline (e.g. New Lead, Not Connected, Interested, Verified, Junk)
First Shipment Booked Date	When the merchant booked their first shipment. Blank if they haven't. This is our definition of a conversion
Marketing Lead Source	Where the lead came from (Organic, Google Paid, Meta, Facebook, App etc.)
Monthly Order Volume	Self-reported volume bucket (Just starting out, 1-100, 100-300, 300-500, 500+)

Tab 3 — Lead Timestamps

Same set of leads, one row each. These are event timestamps — the exact moment an agent moved the lead into a particular stage or acted on it. A blank means the lead never reached that stage.

Created On	Same as Signup Date in Lead_Data
First Phone Call	When the first call was placed to this lead
Latest Follow Up	Timestamp of the agent's most recent follow-up action
Interested	When the agent marked the lead 'Interested'
Offer Pitched	When the agent marked the lead 'Offer Pitched'
Verified	When the lead was marked 'Verified'
Service Done	When the lead reached 'Service Done'

To calculate turnaround times, subtract one timestamp from another. Only use leads where both are filled.

Tab 4 — Call Log

One row per phone call in the last 7 days. This is call-level data, not lead-level, so a lead may appear across several call rows or none. Columns: Start Time, Agent, Call Type (Outbound or Inbound), Duration (in seconds), Disposition (Answer, No Answer, Busy, Failed, Cancel).

A few definitions

- **Conversion:** a lead has a First Shipment Booked Date, or Contact Stage is Verified or Service Done.
- **Working hours** for TAT: Monday to Saturday, 10:00 to 18:30. Sundays excluded.
- **Handoff:** Filtration agents own the lead first. Once qualified, it moves to the paired Sales agent. If a lead's current Owner is a Sales agent, the handoff has already happened.

Task 1 — Funnel Drop-off

From Lead_Data, count how many leads are in each Contact Stage over the last 30 days. Point out the single biggest drop-off in the funnel (where the most leads are stuck or lost). In no more than two sentences, share why you think that's happening.

Task 2 — Segmentation Analysis

Please run the following three analyses. Present each as a small table followed by a one or two line takeaway.

(a) Marketing performance by lead source

For each unique Marketing Lead Source, work out the number of leads, number converted, and conversion rate. Which source is the best performer and which ones would you question the marketing spend on?

(b) Merchant size vs conversion

For each Monthly Order Volume bucket, work out the number of leads and the conversion rate. Does the size of the merchant correlate with conversion? Which segment should the team be spending more time on?

(c) Call connectivity analysis

From Call_Log, work out the percentage share of each Disposition value. What share of calls fail to connect and what does that tell you about how calling is being done?

Task 3 — Additional Insights & Efficiency Memo

Step 1. Look at the data on your own and pull out two more insights that aren't already covered in Task 1 or Task 2. Some angles worth exploring: individual agent performance, response time patterns, data quality issues, or how the funnel is moving over time. Present each one the same way as Task 2: insight, supporting number, and a 'so what' action.

Step 2. Write a one-page memo that pulls everything together (Task 1, Task 2, and your two new insights). Recommend three concrete actions the operations manager should take next week, in the order you'd tackle them. Each recommendation should be specific enough to execute (what, how, by when), backed by a specific number, and worth a manager's time.

What to send back

- The working file (Excel or Google Sheets) with the outputs of Task 1 and Task 2. Please leave formulas visible so we can see how you got to the numbers.
- The one-page memo (PDF or Word).